Mark Jenel Cortas

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SUMMARY

Information Technology student and full-stack developer who builds real-time web applications with TypeScript, React, Next.js, Node.js, Spring Boot, and PostgreSQL. Built and deployed an exam-integrity platform that monitored 100+ concurrent students, and a server-authoritative multiplayer game.

EDUCATION

Polytechnic University of the Philippines | Sta. Mesa, Manila **Expected 2028**
Bachelor of Science in Information Technology

- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Data Communications & Networking, Web Development

TECHNICAL SKILLS

Languages: Java, C, JavaScript, TypeScript, SQL, HTML, CSS
Frameworks & Libraries: React, Next.js, Node.js, Spring Boot, Socket.io, Tailwind CSS
Databases: PostgreSQL, Supabase (Row-Level Security, RPC functions)
Tools & Testing: Git, GitHub, VS Code, IntelliJ IDEA, Vite, Playwright, Chrome Extensions (Manifest V3)
AI-Assisted Development: Claude, Gemini, ChatGPT, Antigravity

RELEVANT EXPERIENCE

AWS User Group Philippines, BuildOverNights Hackathon | BGC, Taguig **Sept. 2026 - Oct. 2026**
Quality Assurance Tester

- Executed end-to-end functional and regression testing on **2 staging web applications**, running **10+ test cases** from standardized runbooks.
- Performed security isolation, offline and throttled network smoke tests, and input validation checks, identifying 3+ **bugs** before release.
- Documented bug reports and QA sign-off checklists on 10+ **GitHub issues** to communicate release readiness to **development team members**.

Google Developer Groups on Campus PUP | Sta. Mesa, Manila **Nov. 2024 - Present**
Operations Associate

- Served as lead host of SparkFest 2025, presenting the online opening ceremony and the on-site hackathon for **100+ attendees** and **20+ teams**.
- Coordinated schedules and live segments for **8 speakers and judges** across online and on-site platforms.
- Co-wrote scripts and planned program flow and logistics for **2 events**.

PROJECTS

CheatNoNo, Exam Integrity & Anti-Cheat Platform | GitHub **Oct 2026**
TypeScript, React, Vite, Chrome Extension (Manifest V3), Supabase, PostgreSQL, Playwright

- Built and deployed a Chrome extension and real-time proctor dashboard that monitored **100+ concurrent students** across **2 live academic exams**, flagging **100+ integrity violations**.
- Secured violation audit logs using PostgreSQL Row-Level Security (RLS) policies and atomic SECURITY DEFINER RPC functions, preventing client-side tampering across **logged events**.
- Developed real-time event-tracking pipelines and an automated integrity-scoring algorithm covering **5+ behavior types**, including tab switches, window blur, copy/paste.
- Automated CSV reconciliation of Google Forms submissions against live telemetry and departure heatmaps, cutting proctor review time from **10 minutes to 20 seconds** per exam.

Countdown Math Arena | GitHub **Oct 2026**
Next.js, TypeScript, Socket.io, Node.js, Supabase, PostgreSQL, Tailwind CSS

- Designed a server-authoritative real-time multiplayer game on a dedicated Node.js and Socket.io server, synchronizing match state, round timers, and reconnects for 2 **players.**
- Built an expression validation and scoring engine with recursive solution checking and strict arithmetic-rule enforcement, rejecting invalid and late submissions in timed rounds.
- Implemented an atomic, idempotent PostgreSQL match-finalization function that updates MMR, wins, and losses, preventing duplicate stat updates across **100+ matches** played by **10+ users**.